



Performance evaluation of solar-combined boosting topology for EV battery charger using interval type-2 fuzzy controller

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Abstract

This paper offers a unique and novel approach for electric vehicle battery charging by incorporating a solar PV-integrated dc to dc boost converter. Traditional power electronic converters used in such applications often face limitations including low voltage gain, low conversion efficiency, high ripple content, and inadequate controller performance. To overcome these challenges, the study introduces the Improved A-Source Boost Converter (IASBC) along with an interval type-2 fuzzy logic controller (IT2FLC) and MPPT technique for continuous switching with optimal design. The IASBC, equipped with a coupled inductor, enhances load side performance by achieving high conversion efficiency and voltage gain while minimizing the impact of network parasitic. Additionally, the proposed controller technique effectively reduces output side dc ripple content through gain parameter optimization. The system is optimized for charging a 48 V rechargeable battery within a solar photovoltaic voltage range of 25–75 V DC. Extensive simulations using MATLAB validate the efficacy of the proposed approach. The software analysis reveals an efficiency of 90.82% in the open-loop system and 98.57% in the closed-loop system, while the hardware system achieves an efficiency of approximately 98.55%. Furthermore, a 100 W experimental design is developed and compared with the simulation results, providing validation for the technical significance of the suggested converter. This novel approach not only addresses the limitations of conventional converters but also offers improved performance in terms of efficiency, voltage gain, and ripple content. The proposed system holds promise for efficient electric vehicle battery charging and can contribute to the advancement of sustainable energy solutions.

Keywords Photovoltaic · Output gainer · Battery load · IT2FLC · Maximum power point tracking

1 Introduction

Green energy plays an essential part in today's electric power conversion systems. The project's goal is to develop a highly reliable energy technique that can work with solar and wind power. This study examined multiple boosting, buck, and buck-boost network topologies using various control methodologies to demonstrate their technological distinctiveness. Most of the research describes the use of multi-objective adaptive algorithm-based regulators to eliminate technical factors such as ripple voltage and current waveforms at parasites in the network. As a result, the system's dependability and performance may be improved. A revolutionary distributed generating system for a hybrid energy storage system, based on a Phenomenon Optimization algorithm, with a topology that incorporates Z-source conversions into medium voltage DC grids. Using the adaptive Biogeography-Based Optimization approach, the energy management system (EMS) optimizes an objective function

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characterizing similar hydrogen utilization. The charging station consists of a solar system, an electrical network link, two fast-charging units, two energy storage systems, a battery energy storage, and a complete hydrogen system with a fuel cell, electrolyzer, and gasoline tank [1]. The fundamental source of the problem is poor capacitive reactance selection. To address such challenges, multi-objective issues such as control system input signals and capacitor selection are performed. For optimization, the NSGA-II and extended NSGA-II algorithms are utilized. In this study, the Global optimization approach is applied as a power converter to give maximum tuning for a Control system. The charge controller is represented mathematically by the formulation, which is then modeled in MATLAB/Simulink. The proposed Gray Wolf Optimization-PID tweaked model is equivalent to Particles Swarm Optimization (PSO) and Genetic Algorithms (GA) techniques in terms of Root Mean Squared Errors (RMSE) and system responsiveness for diverse loads and shifting voltage levels. The constant and irregular conducting operation modes of the single switched impedance network (SSIN)-based converters are addressed. The influence of parasite components on the SSIN's Converter gain and performance is investigated. The SSIN converter's comparatively tiny theory is obtained. Finally, the SSIN converter's method is measured to comparable power converters [2]. There is still a need for a theory or fundamental construction piece that allows for integrated synthesizing, evaluation, projection of output currents, and switching state assignments across a large range of DC conversion. Three 3-terminal simple construction elements and one three-terminal filtering element will be proposed in this research. It will be demonstrated that they are adequate for implementing all non-isolated DC-DC converters, excluding those with connected inductances, when used together [3]. The duty ratio of the high-voltage component is exclusively determined by asymmetrical pulse width modulation, whereas the LV active-bridge functions with a constant duty ratio. The key component appears to be achieving zero voltage and zero current switching operations throughout a wide power spectrum with only one parameter regulating the converters. The a-PWM efficiently removes current circulation, and antiparallel diodes were used to avoid any current flow reversal [4].

PSO and Differential Evaluation are two traditional evolutionary techniques that may enhance the composite MPPT technique. The suggested hybrid MPPT approach's major characteristic is that the benefit of one approach substitutes for the inadequacies. Moreover, the method is straightforward and quick. It is simple to put together a reduced microprocessor [5]; this paper provides a model predictive control (MPC)-based MPPT approach using two parameters with the goal of lowering costs. First, the method supports static or adjustable step sizes—a multi-objective

intended goal for maximizing economic and social advantages in the metal mines production process (MMPP). An enhanced non-dominating sorting genetic procedure II is presented to acquire improved non-dominated Pareto optimum solutions. Finally, a comprehensive co-design optimization framework for multiphase SiC-based DC/DC converter uses a dual-loop control scheme to promote an efficient energy conversion phase and a regulator stage [6, 7]. Regarding the converter's Continuous mode, a voltage equalization component architecture is added, which might establish voltage balancing between the masters and slave converters [8]. A digital prototype obtained on an optimization technique and simulation studies is built, and it is then utilized to improve the converters. The proposed method seeks to increase conversion performance by reducing transformer size. The generated Pareto front is used to build models. Scientific results demonstrate the reliability of the predicted findings. The findings of the optimal design reveal that switching frequency allows for a significant volume decrease of the inductive elements of the modular multilevel converter (MMC)/LCL filter with minimal effect on the MMC's power dissipation. The Z-source inversion is a renowned one-stage power conversion scheme instead of a normal voltage source and current source inverters. It improves input voltage dependability by adding an extra buck-boost capacity [9–11]. In the A-source impedance network, the auto-transformer is utilized to realize any activity that requires a very high-voltage level gain. To maintain a high-voltage output, the circuit has a low duty ratio when compared to current Power Converter Impedance Source networks. Furthermore, because the recommended converters extract a current source from the supply, it is appropriate for a variety of sustainable energy sources [12]. A Phase shifted full bridge (PSFB) Power converter has a minimum dc output of 12 V and is valued for a rated power of 2 kW. The primary goal of this study is to identify Controller gains settings that keep the settling time, steady-state error, and peak overshoot inside a prescribed range for all technology in the process [13]. The whole control network structure for the voltage-mode regulated two-phase Interleaved Charge controller was created utilizing a simple analog circuit design, and the experimental fabrication setup was inexpensive [14]. The suggested converters have reduced inductance losses because the decreased load current ripple reduces the maximum inductor current, but the converters have more passive components [15]. The output voltage control-loop control algorithm is an optimization method, and the regulator constant is found through adaptive search. The design approach, which includes evolutionary algorithms, differential evolution, and an artificial immune response, is discussed [16, 17]. This paper presents mathematical optimization methodologies for getting network component values for buck converters in order to decrease output voltage variation during load shifting [18,

19] The dc-dc converters, primarily used to maximize the output produced by the PV array, are not included in the systems. Furthermore, storage devices are not enclosed [20, 21].

The stability verification of soft-computer-based solutions can benefit from utilizing the TP (Tensor product) model transformation as a comprehensive framework. In addressing the criticisms often directed at fuzzy or soft-computer-based control solutions, the TP transformation presents a promising approach [22]. By constructing a fuzzy membership-dependent Lyapunov function, sufficient conditions can be derived in the form of solvable linear matrix inequalities (LMIs). These conditions facilitate the determination of suitable control gain matrices, ensuring the stabilization of wind energy conversion systems [23]. For representing nonlinear systems with intricate uncertainties, the proposal of interval type-2 (IT2) fuzzy models [24] offers a viable solution. These models are constructed by incorporating type-2 fuzzy sets, thereby accommodating uncertain parameters or unknown variables within the membership functions of the fuzzy models.

The application of this modeling approach has significantly expanded its usage in uncertain systems. Furthermore, empirical evidence has demonstrated that the IT2 fuzzy model, based on type-1 fuzzy sets, offers a less conservative alternative. A novel strategy for the neutral-point-clamped converter is proposed, utilizing a fuzzy logic system-based sliding-mode control. To ensure convergence of tracking errors within bounded regions, an efficient super-twisting algorithm based on a fuzzy logic system is employed in the power tracking loop [25].

To enhance the MPPT technique, an optimization process is introduced by adjusting the membership function according to the converter's topology and switching frequency [26]. With 12 rules, the fuzzy set governs the decision-making process. During the defuzzification phase, the centroid method is utilized to calculate the center of gravity (COG) within a specific membership function area. By averaging in the centroid method, the control action is smoothed, rendering the controller less susceptible to minor fluctuations [27].

As is widely acknowledged, the FCS-MPC demonstrates characteristics of being a control approach based on models, making it highly reliant on system parameters and external disturbances [28]. It is important to note that the presence of parametric uncertainties and unknown system dynamics introduces a bias error between the reference and actual controlled variables. Consequently, the future evolution of system variables under each potential control action may follow an undesirable trajectory, inevitably impacting the system's performance. In order to address these challenges, a parametric optimization outline is proposed for the mixture of a strong adaptive fuzzy regulator designed for a specific stage of triggering power converters. The focus lies on open-looping systems, specifically phase-shifting pulse width

modulation full-bridge DC-DC converters, which possess applied advantages like extensive-range voltage output and higher efficiency. To facilitate subsequent optimization and simulation processes, a comprehensive mathematical model, alongside its corresponding numerical representation, has been established for this converter in prior work [29].

A novel WFS (wire feeder system) configuration is proposed, presenting an alternative approach to energy systems by integrating solar PV technology. Specifically, the focus is on developing a standalone solar PV-WFS system aimed at maximizing power output from a PV generator and improving welding quality. To achieve this, the implementation of an FO-Fuzzy PID regulator is recommended. Solar energy is widely recognized as a sustainable and environmentally friendly source of power [30]. However, the energy generated by a PV generator is subject to variations due to weather conditions. Therefore, it becomes crucial to employ MPPT methods to yield the supreme energy from the PV generator under diverse temperatures and radiation levels. To effectively handle the inherent uncertainties, an Interval Type-2 Fuzzy Logic Control (IT2FLC) strategy is adopted. IT2 Mamdani FLCs have been extensively studied for a range of industrial applications, including speed control of marine diesel engines, liquid level control, temperature control of steel strip mills, and mobile robot control [31, 32]. Through the proposed PV-integrated WFS system and the utilization of an FO-Fuzzy PID regulator, it is expected to achieve enhanced power optimization and welding quality, thereby advancing the field of wire feeder systems in welding applications.

The summary of literature mentioned in Table 1 clearly explains that there are various power converters are designed and validated using different controllers. But still, those controllers are lagging in various aspects, such as ripples at the output side, higher convergence rates, and large mean square error values. The above drawbacks could be eliminated by the proposed topology with the proper design of power converters and optimization techniques for PI tuning purposes.

As a result of the literature, it is clear that when combining photovoltaic Panels with a power conversion for recharging a rechargeable battery, the difficulties caused by partial shade, including reduced voltage gain and voltage drop, are unpleasant. Therefore, an appropriate Control scheme with improved gain parameters improves system results. However, the traditional boosting circuit with Controller parameters optimization design produces more voltage ripple at the filter output, fast switching losses, and thus reduced system efficiency, which is recognized as a multi-objective problem. Therefore, a determination is taken on the complete system by integrating the IASBC with MPPT and T1FLC / IT2FLC algorithms that improve load side operation with high conversion efficacy and high-voltages gain due to the effect of a coupling inductor and minimize the impacts of network parasitic.

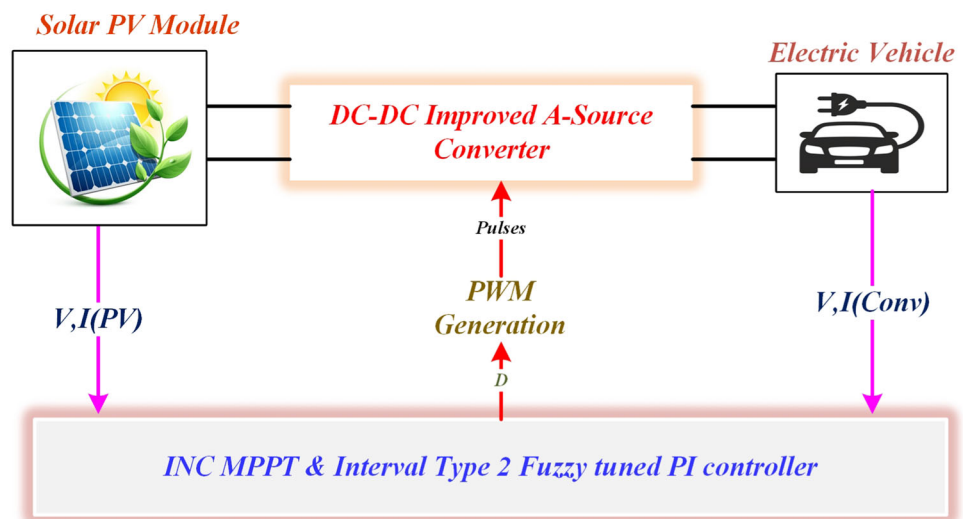
Table 1 Literature review summary

Authors	Scheme	Arrangement	Particular
[33]	Optimizations built dc-dc converters	Existing Boost Converters	The comparison involved assessing the convergence and iteration of PID parameters using different techniques like GA, PSO, and GWO
[34]	Power converters utilizing evolutionary algorithms as the basis	Buck Converters	To enhance the overall system performance, a multi-objective approach is employed for fine-tuning the PID gain values
[35]	A PI controller is implemented in solar-combined DC-DC converters	Modified A-source converter	Introducing a PI controller based on MPPT enhances the performance of the load
[36]	Innovative DC-DC Converters for PV Systems	Cuk-SEPIC Converter	As a bipolar DC link, this solution is well-suited for PV systems
[37]	Differential Evolutions-based MPPT Algorithms	SEPIC converter	The system demonstrates a remarkable capability to track and swiftly respond to load variations
[38]	NSGA-II implemented power converters	Interleaved boosting converter	Three key objectives, namely minimizing total input current ripple, reducing the total weight of inductors, and minimizing total power losses, have been selected for this multi-objective analysis

The output side dc ripple content is reduced by the optimal gain parameters produced through the proposed optimization technique. The SPV combined IASBC converters with MPPT and IT2FLC-based controller technique are presented to solve the preceding challenges, with the accompanying goal in mind:

- Design and implement a solar PV incorporated dc to dc boost converter for electric vehicle battery charging applications.
- Overcome the limitations of conventional power electronic converters, including low voltage gain, low conversion efficiency, high ripple content, and poor controller performance.
- Introduce the Improved A-Source Boost Converter (IASBC) with a coupled inductor to enhance load side performance, achieving high conversion efficiency and voltage gain.
- Incorporate the MPPT and (IT2FLC) technique for continuous switching with optimal design, improving the controller performance of the boost converter.
- Optimize the gain parameters of the boost converter using a proposed optimization technique to minimize output side dc ripple content.
- Optimize the system for charging a 48 V rechargeable battery, considering the solar photovoltaic voltage range of 25 to 75 V DC.
- Validate the proposed approach through simulation using MATLAB, assessing the performance, efficiency, and stability of the charging system under various operating conditions.
- Develop a 100W experimental design and compare its technical performance with the simulation results, validating the effectiveness and feasibility of the proposed converter in practical applications.
- Contribute to the advancement of electric vehicle battery charging technology by providing an improved and optimized solution that addresses technical limitations and enhances the efficiency and reliability of the charging process.
- Promote the adoption of renewable energy sources, such as solar PV, for electric vehicle charging, contributing to the sustainable development of transportation systems.

The proposed effort is systematized in a subsequent manner: Sect. 2 explains the proposed techniques with their mode operations, switching arrangements, and analytical formulae. Section 3 presents the proposed methodology's Prominence of the IT2FLC controller strategy, including its control architecture of the proposed controller and its design considerations along with the stability model of the proposed converter. Section 4 then shows the Closed-Loop Implementation of the Proposed Topology using MATLAB Simulink

Fig. 1 Architecture of the Proposed Methodology

software which further includes Pre-set values of the proposed controller types and MATLAB Simulation outputs and an investigation of the proposed converter. Section 5 provides the Experimental Outfit and Results of IASBC. Finally, Sect. 6 ultimately exposes the conclusion.

2 Proposed methodology

Figure 1 shows the proposed converter's design, in which solar PV is used to charge the EV battery via an IASBC converter. The switches are triggered by the PI controller generating an error signal by measuring the measured and reference values. The MPPT and IT2FLC controllers are used to tune the controller gain values. A power amplifier utilizing pulse width modulation is employed to receive a low-power input from the controller board and deliver a high-current driving input to the gate of a power MOSFET. Both simulation and hardware setups have been developed for the proposed topology.

2.1 Circuit model of the proposed converters

Figure 2 illustrates the configuration of the proposed converter, presenting a closed-loop system design. The key components depicted in the diagram include photovoltaic panels, an IASBC (Integrated Active and Standby Battery Charger), MPPT (Maximum Power Point Tracking) and IT2FLC (Interval Type-2 Fuzzy Logic Controller) controllers, as well as RC loads representing the battery impedance.

The IASBC, an integral part of the converter, is comprised of several elements. These elements consist of a controlled switch denoted as S , an inductance denoted as L , and two capacitors designated as C_A and C_B . Additionally,

the IASBC incorporates an auto-transformer that features primary and secondary turns, represented, respectively, by N_A and N_B . Furthermore, the auto-transformer comprises primary and secondary inductances denoted as L_A and L_B .

The highest output voltage of the Improved A-Source circuits is V_o . According to the need, a filtering circuit could be put at the points of V_o to create dc-dc conversions. The A-Source circuit converts the energy between dc-dc. The MPPT and IT2FLC controllers create a pulse to activate the switch S and give the load the most significant amount of power.

2.2 Modes of operations

Figure 3a depicts the Improved A-Source network's equivalent circuits in the condition of shoot-through arises. The switch S is open and the diodes D_A is off throughout the shoot-through condition. Whenever we use KVL to Fig. 3a, we obtain the following Eq. (1) [12],

$$V_i - V_L + V_{CB} + \frac{N_B}{N_A} V_{LA} + V_{LA} - V_{CA} = 0 \quad (1)$$

We know that from Fig. 3a,

$$V_{LA} - V_{CA} = 0 \quad (2)$$

By rearranging (1), we get

$$V_L = V_i + V_{CB} + \left(\frac{N_B}{N_A} + 1 \right) V_{LA} - V_{CA} = 0 \quad (3)$$

The voltage across the inductors throughout the triggering condition is evaluated using (2) and (3).

$$V_L = V_i + V_{CB} + \frac{N_B}{N_A} V_{CA} \quad (4)$$

Fig. 2 Circuit model of the proposed topology

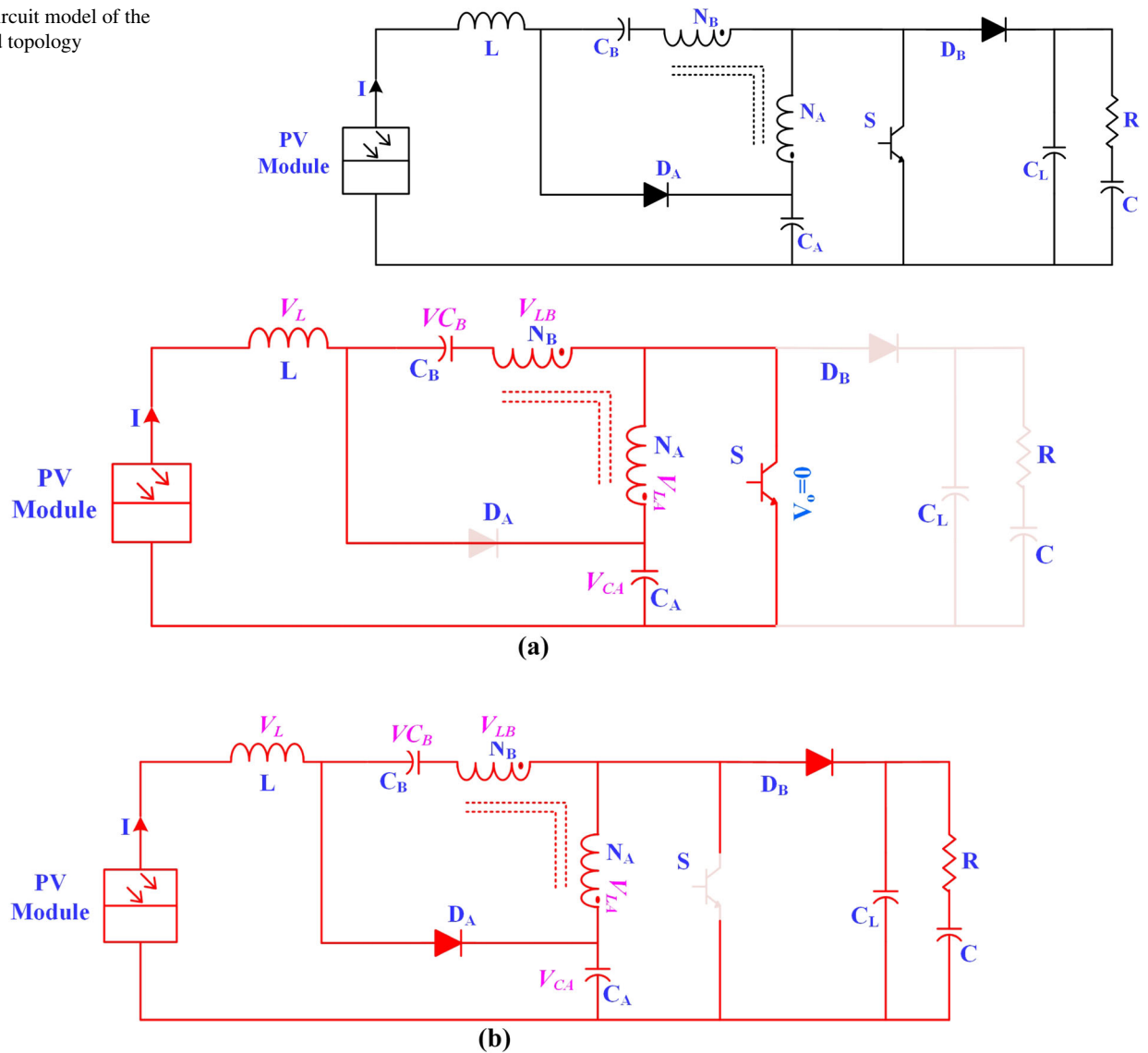


Fig. 3 Modes of Operation of the proposed converter (a) Mode1 (b) Mode 2

Figure 3b represents the IASBC equivalent circuitry in the non-shoot-through condition. The switch S is turned off, and the diodes D_A is forward-biased. The equations of states are as follows:

$$V_L = V_i - V_{CA} \quad (5)$$

To find V_{LA} , Eq. (6) is written as follows:

$$V_{CB} + \frac{N_B}{N_A} V_{LA} + V_{LA} = 0 \quad (6)$$

$$V_{LA} = -\frac{V_{CB}}{\frac{N_B}{N_A} + 1}$$

We may use the volt-second balancing concept to balance the inductor L_A .

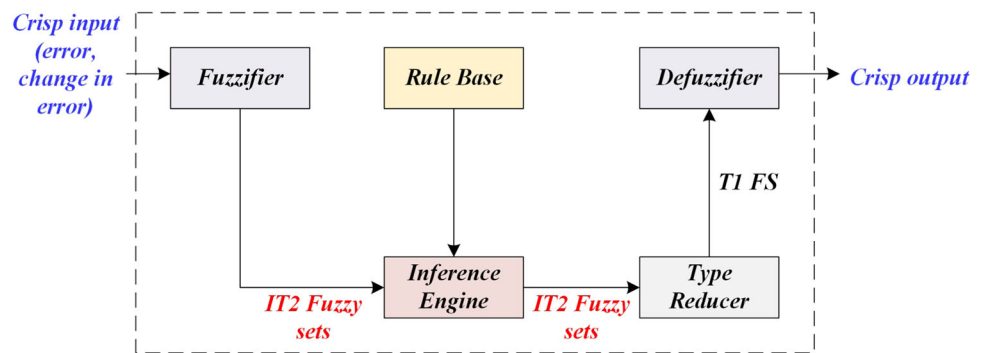
$$\int_0^{d_{st}T_s} V_{LA} dt + \int_{d_{st}T_s}^{T_s} V_{LA} dt = 0 \quad (8)$$

When (2) and (7) are substituted for (8), the result is

$$V_{CA} d_{st} - \frac{V_{CB}}{\frac{N_B}{N_A} + 1} (1 - d_{st}) = 0 \quad (9)$$

Yielding,

$$\frac{V_{CA}}{V_{CB}} = \frac{1 - d_{st}}{\left(\frac{N_B}{N_A} + 1\right) d_{st}} \quad (10)$$

Fig. 4 Configuration of IT2FLC

where d_{st} denotes the switch's shoot-through duty cycle. When the volt-second balancing principle is applied to the inductor L , the result is

$$\int_0^{d_{st}T_s} V_L dt + \int_{d_{st}T_s}^{T_s} V_L dt = 0 \quad (11)$$

Equations (3) and (5) are replaced with (11) to produce

$$\left(V_i + V_{CB} + \frac{N_B}{N_A} V_{CA} \right) d_{st} + (V_i - V_{CA})(1 - d_{st}) = 0 \quad (12)$$

$$V_i + \left(\frac{N_B}{N_A} d_{st} + d_{st} - 1 \right) V_{CA} + d_{st} V_{CB} = 0 \quad (13)$$

The voltage between the capacitors C_A and C_B maybe determined by solving (10) and (13).

$$V_{CA} = \frac{1 - d_{st}}{1 - \left(2 + \frac{N_B}{N_A} \right) d_{st}} = \frac{1 - d_{st}}{1 - (1 + N) d_{st}} V_i \quad (14)$$

$$V_{CB} = \frac{\left(1 + \frac{N_B}{N_A} \right) d_{st}}{1 - \left(2 + \frac{N_B}{N_A} \right) d_{st}} = \frac{N d_{st}}{1 - (1 + N) d_{st}} V_i \quad (15)$$

where $N = N_A + N_B / N_A$ is the auto-transformers turns value.

As a result, the voltages transfer ratio G is required to provide an output voltage of V_o during the non-shoot through time is determined using Eq. (16),

$$G = \frac{V_o}{V_i} = \frac{1}{1 - (1 + N) d_{st}} \quad (16)$$

The input-to-output DC transfer characteristic is non-shoot through, assuming a 100 percent efficiency, thereby resulting in the following consequences,

$$G = \frac{I_{om}}{I} = \frac{1}{G_v} = 1 - (1 + N) d_{st} \quad (17)$$

3 Description of IT2FLC-based control strategy for a power solution

Within this section, the focus lies on the control circuit design of the IASBC converter employing a proposed controller. Figure 4 illustrates the standard block diagram representing the proposed IT2FLC. The structure of the IT2FLC consists of a fuzzifier, a rule base, and an inference engine, similar to the conventional T1FLC. However, it further incorporates a type reducer and a defuzzifier within the output processing unit, distinguishing it from the traditional T1FLC.

3.1 Control architecture of the IASBC converter based on IT2-FLC

Figure 5 depicts the illustrated control configuration of the IASBC converter utilizing IT2FLC in a closed-loop fashion. This control methodology implements both an external voltage control loop and an internal current loop, responsible for regulating the desired DC output voltage and output current limitations. Initially, the voltage error (V_E) is generated by comparing the reference voltage (V_0^*) with the sensed actual output voltage (V_0). Simultaneously, a delay unit is integrated to calculate the variation in the error signal (V_{CE}) at the n -th sampling time instance. The mathematical representation of the voltage error signal can be expressed as follows:

$$V_E(m) = V_0^* - V_0$$

$$V_{CE}(m) = V_E(m) - V_E(m - 1) \quad (18)$$

The voltage controller is responsible for processing the output error signal, while the internal units handle the adjustment of the current reference signal (ΔI_0^*) to generate the desired current reference signal (I_0^*). Simultaneously, the error signal (I_E) is formed by comparing the current reference signal (I_0^*) with the sensed output current (I_0). In a manner similar to before, the change in error signal (I_{CE}) at the n -th sampling instant is obtained using a delay unit. Thus,

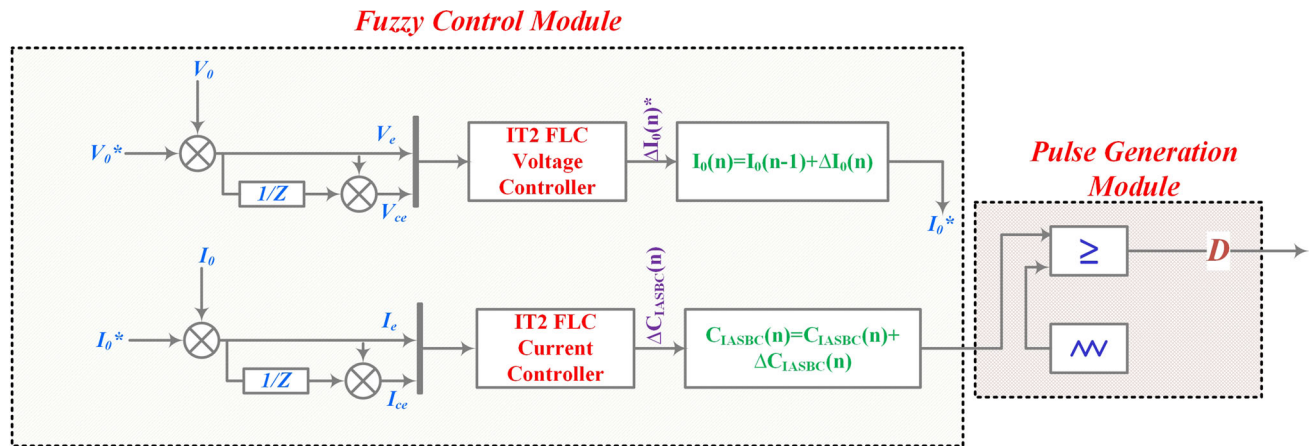


Fig. 5 Visual depiction illustrating the power solution modeled using IT2FLC

the current error signal can be mathematically expressed as follows:

$$I_E(m) = I_0^* - I_0$$

$$I_{CE}(m) = I_E(m) - I_E(m-1) \quad (19)$$

Moreover, the primary control signal C_{IASBC} is derived from the variation in the control signal ΔC_{IASBC} , obtained from the output of the current controller based on IT2FLC. This process involves the utilization of internal delay units. The change in the control signal C_{IASBC} at the n -th instant is then defined as follows:

$$C_{IASBC}(m) = C_{IASBC}(m-1) + \delta C_{IASBC}(m) \quad (20)$$

Afterward, the control signal C_{IASBC} is fed into the PWM generator, which generates the necessary firing pulses for the power switches (S) to achieve the desired outcome.

3.2 Design of IT2FLC for power solution based on the IASBC converter

To capture uncertainties in linguistic variables and reduce computational complexity, IT2FSs were created. This study adopts the Takagi–Sugeno–Kang fuzzy inference model for computations within IT2FLC, ensuring minimal computational burden. Conventionally, the type-2 fuzzy set is denoted as A and can be represented as follows:

$$X = \{(y, v), \mu_X(y, v)\}, \forall y \in Y, \forall v \in K_y \subseteq [0, 1] \quad (21)$$

wherein, $X = u \in Y, v \in K_y[0, 1] 0 \leq \mu_X(y, v) \leq 1$

With the objective of mitigating computational complexities, an IT2FS denoted as $\tilde{X}$ is introduced, and its mathematical representation is given by:

$$\tilde{X} = \int_{y \in Y} \int_{v \in K_y} \frac{1}{(y, v)}, K_y \subseteq [0, 1];$$

$$\tilde{X} : Y \rightarrow \{[p, q] : 0 \leq p \leq q \leq 1\} \quad (22)$$

Typically, the IT2FLC framework consists of upper and lower membership functions (LMFs and UMFs). The area bounded by two fuzzy sets is referred to as the FOU (Fuzzy Output Universe), which defines the range of ambiguity.

$$FOU(\tilde{X}) = \bigvee_{y \in Y} Y K_y = \{(y, v) : v \in K_y \subseteq [0, 1]\} \quad (23)$$

A visual representation of the embedded fuzzy set $\tilde{X}_E$ within the universe of discourse Y is depicted.

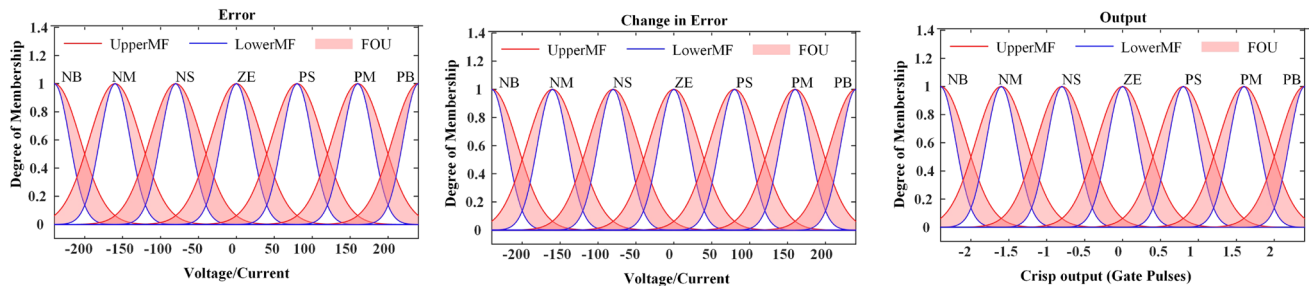
$$\tilde{X}_E = \int_{y \in Y} \left\{ \frac{1}{v} \right\}_y, v \in K_y \quad (24)$$

The following steps outline the design process for the preferred power solution based on IT2FLC:

- **Fuzzification:** The first step involves mapping crisp inputs to fuzzy sets using UMF and LMF. The inputs, error (E), change in error (CE), and output, ΔC_{IASBC} , are associated with SEVEN membership functions: Big Negative (B^-), Meager Negative (M^-), Slight Negative (S^-), Zero (z), Slight Positive (S^+), Meager Positive (M^+), and Big Positive (B^+). The universe of discourse for error (E) and change in error (CE) ranges from -1 to 1.
- **IF–THEN Rule Formation:** IF–THEN rules are formulated to create the rule base. For example, IF 'E' is nh and 'CE' is NH, THEN 'z = AE + BCE + C'. Here, nh represents the respective mfs of the inputs E and CE, while A, B, and C are attributes of the TSK inference model. Similar rules are created for other mfs and are tabulated in Table 2, resulting in a total of 25 rules.

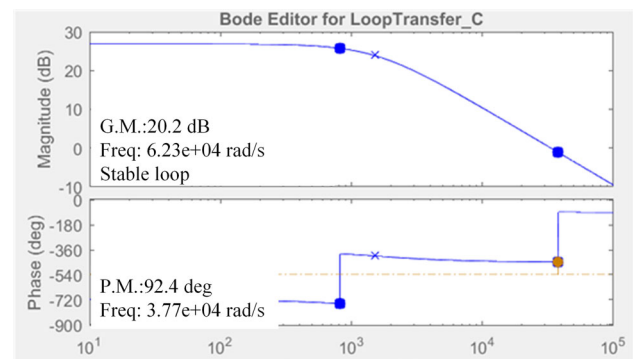
Table 2 Rules of IT2FLC

	Change in error (CE)	Error(E)						
		B ⁻	M ⁻	S ⁻	Z	S ⁺	M ⁺	B ⁺
S ⁻		B ⁻	B ⁻	B ⁻	M ⁻	M ⁻	S ⁻	Z
M ⁻		B ⁻	M ⁻	M ⁻	S ⁻	S ⁻	Z	S ⁺
B ⁻		B ⁻	M ⁻	S ⁻	S ⁻	Z	S ⁺	M ⁺
Z		M ⁻	S ⁻	S ⁻	Z	S ⁺	S ⁺	M ⁺
B ⁺		M ⁻	S ⁻	Z	S ⁺	S ⁺	M ⁺	B ⁺
M ⁺		S ⁻	Z	S ⁺	S ⁺	M ⁺	M ⁺	B ⁺
S ⁺		Z	S ⁺	M ⁺	M ⁺	B ⁺	B ⁺	B ⁺

**Fig. 6** Membership functions of IT2FLC**Table 3** Considerations for simulation

Power rating	(300–350)W
Modules specification	Sunarray-S6B3612-350 T
Ir(irradiance) W/m ²	1000 @ 25 °C
Output voltages (V _{oc})	25 V DC
Output current (I _{sc})	14 A DC
Inductance L _A , L _B	L _A =45μH, L _B =30μH
Capacitance C _A , C _B	470μF, 3600μF
IASBC Output Voltage	75 V DC
IASBC Output Current	4.67A DC
Duty ratio	d _{st} =0.2; N _A : N _B = 20 : 20; N = 2
R, C	8.23 Ω, 100 μF
K _p , K _i , K _d	0.4, 3.6, 0.001
C filter	C _L = 800 μF
Battery nominal voltage	12 V DC
Capacity	7Ah

- **Fuzzy Reasoning:** The fired rules are combined using a fuzzy reasoning strategy to map the input IT2FS to the output IT2FS. The rule membership functions are visualized in Fig. 6.
- **Type Reduction:** Type reduction methods are applied to reduce computational overhead. This work employs the

**Fig. 7** Simulated graph of the frequency response of PI controller

center-of-sets approach, which utilizes the Karnik–Mendel algorithm to determine the left (Y_L) and right (Y_R) endpoints of the type-reduced set.

- **Defuzzification:** Finally, defuzzification methods are applied to obtain a crisp output from the type reducer. Averaging both the left (Y_L) and right (Y_R) endpoints yield a crisp output.

These steps collectively contribute to the design of the IT2FLC-based preferred power solution.

Fig. 8 PI controller operation diagram with T1FLC / IT2FLC optimization

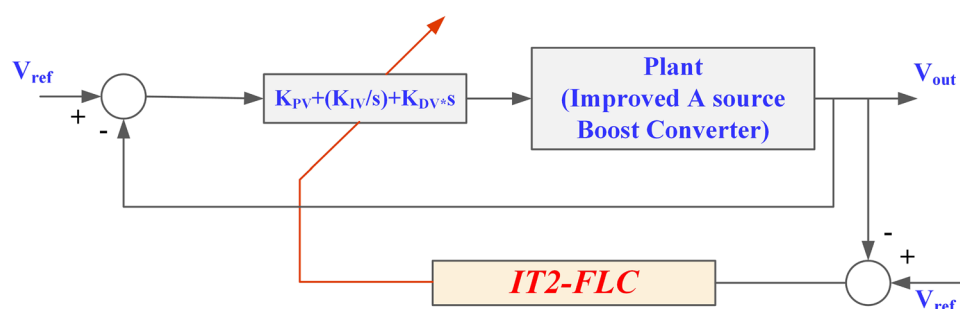


Table 4 Performance of the suggested topology is compared using both established and newly proposed algorithms

Algorithm type	K_p	K_i	K_d	Rise time (ms)	Steady-state error (V)	Settling time (ms)	Peak overshoot (V)	Standard Deviation	Efficiency @ 1000 W/m ² (%)
ZN-PI	0.5	4.2	0.002	0.02	0.4	0.5	1.2	–	91.23
Adaptive Chebyshev Neural Network	–	–	–	–	2	5	13.6	–	< 90
Genetic Algorithm [13]	4.9	4667	–	–	7	3.714	0	0.0095	91
PSO [14]	–	–	–	10	–	20	3.33	0.0098	88
T1FLC	4.92	18.52	0.04	0.04	0.06	0.38	1.06	0.00713	95.72
IT2FLC (Proposed)	8.2	28.63	0.02	0.05	0.01	0.25	1	0.00163	98.57

Fig. 9 T1 FLC outcomes (a) voltage and (b) current

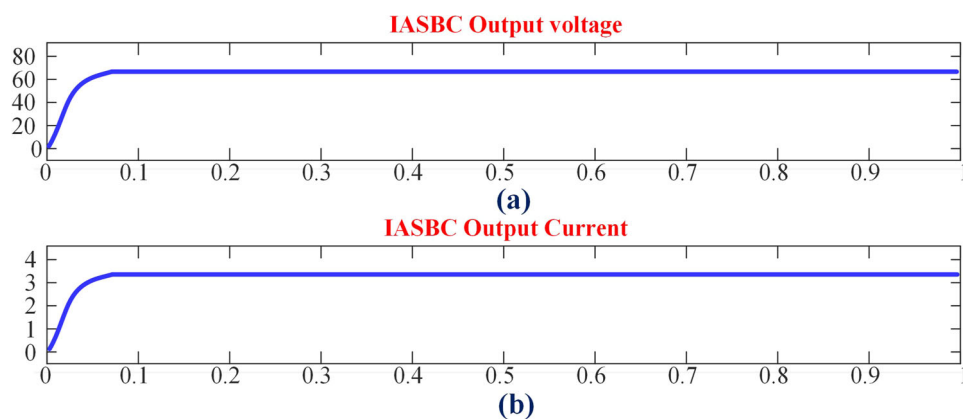


Fig. 10 Simulated ripple output voltages of T1FLC

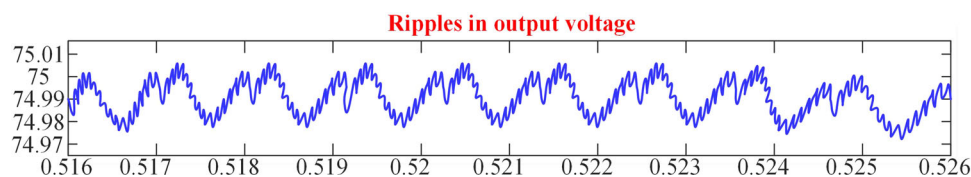


Fig. 11 IT2 FLC outcomes
a voltage and b current

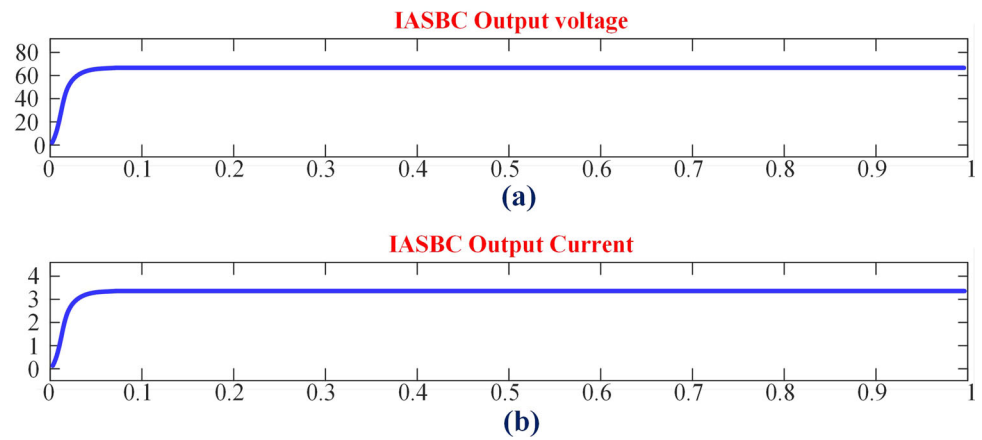


Fig. 12 Efficiency comparison
chart with different controllers

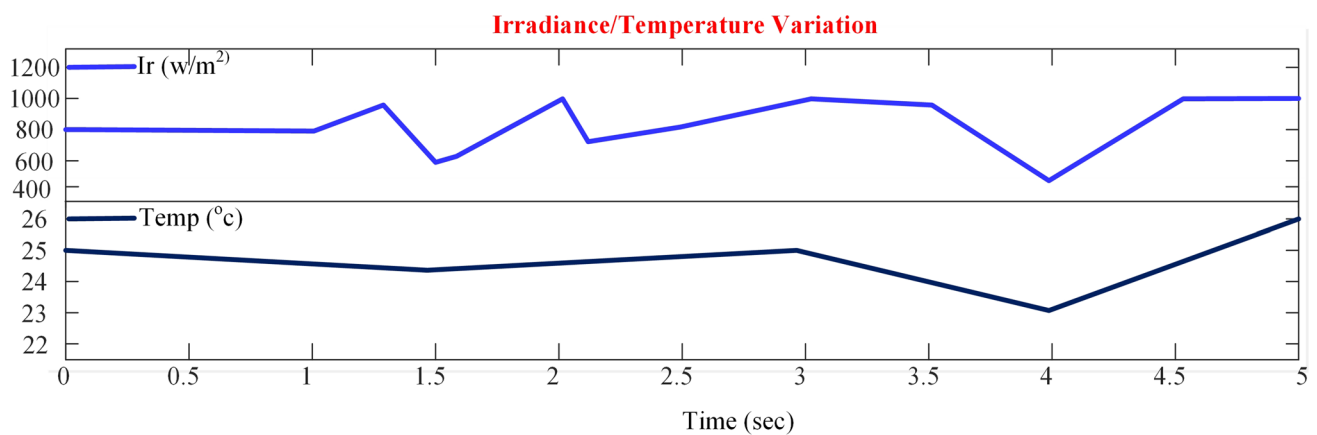
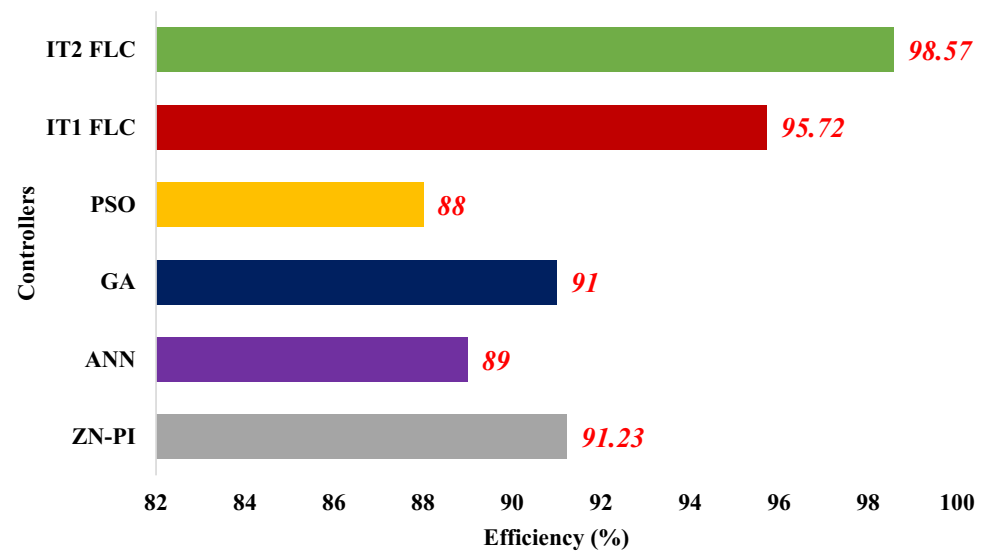


Fig. 13 Irradiation and Temperature graph

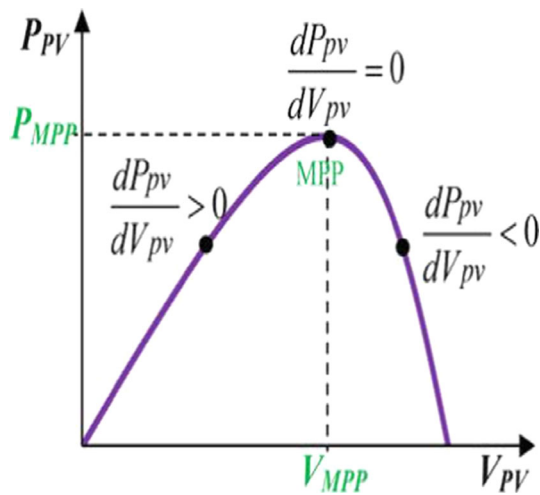


Fig. 14 Incremental Conductance Technique flow chart

4 Considerations of the suggested converter

4.1 Modeling the proposed converter in state space representation

The state space representation is employed to model the proposed converter, providing a comprehensive description of its dynamic behavior. This modeling approach allows us to analyze the system's internal states and their evolution over time. In the state space representation, the converter's behavior is described by a set of state variables, input variables, and output variables. The state variables represent the internal states of the converter, while the input variables correspond to the control signals applied to the system. The output variables reflect the measurable quantities or desired performance of the converter. By formulating the converter's dynamics in state space form, it becomes possible to design controllers, analyze stability, and investigate system responses. The state equations capture the relationships between the derivatives of the state variables and the input variables, while the output equations relate the output variables to the state variables. The state space modeling of the proposed converter provides a powerful framework for understanding and optimizing its operation. It enables the exploration of various control strategies and facilitates the analysis of the converter's performance under different operating conditions.

The modeling of the preferred converter is based on two IASBC modules that operate symmetrically. For simplicity, we can focus on a single module for the purpose of modeling. To accurately represent the circuit behavior, we employ a state space approach that accounts for non-idealities. Utilizing the expressions (1–24) mentioned above, we construct the model of the preferred converter using state variables associated with specific components. These variables include the current flowing through the input inductors (i_L), the voltage

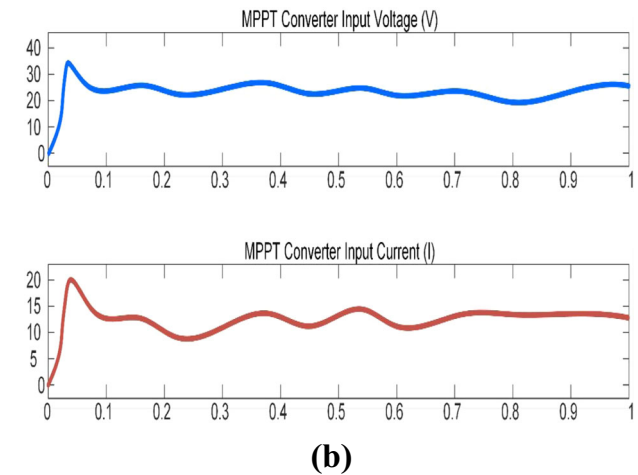
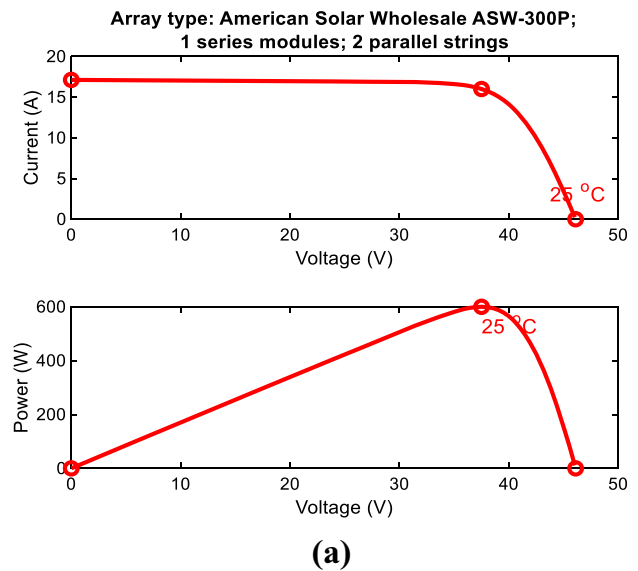


Fig. 15 a IV and PV Curve b MPPT Converter Input Voltage and Current

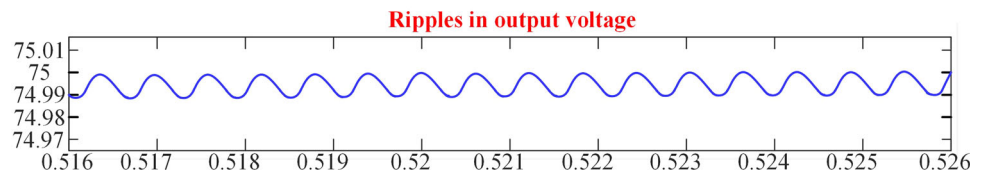
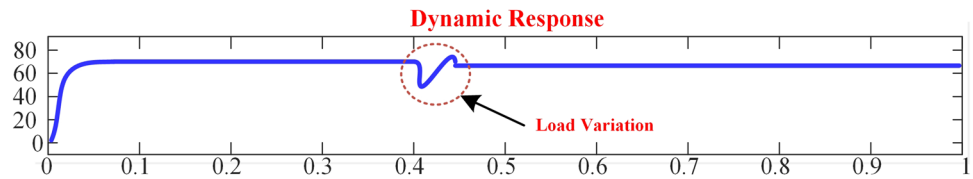
across the intermediate capacitors (v_{C_A} , and v_{C_B}), the current through the intermediate inductors (i_{L_A} , and i_{L_B}), and the voltage across the output capacitor (v_{C_L}). The state equation and output equation corresponding to the model are as follows.

This comprehensive state space representation enables a detailed analysis of the converter's behavior, accounting for the effects of non-idealities and providing a solid foundation for further investigation and control design.

$$\dot{X}_{IASBC} = A_{IASBC} X_{IASBC} + B_{IASBC} V \quad (25)$$

$$V_{Refi} = C_{IASBC} X_{IASBC} \quad (26)$$

$$A_{IASBC} = (A_{IASBC1} * d) + (A_{IASBC2} * d_1) \quad (27)$$

Fig. 16 Simulated ripple output voltages of IT2FLC**Fig. 17** Dynamic behavior of the proposed converter**Table 5** Variations in Irradiance during Open-Loop Analysis

Irradiation (W/m ²)	Duty cycle	v_{in} (V)	i_{in} (A)	P_{in} (W)	v_{out} (V)	i_{out} (A)	P_{out} (W)	η (%)
800	0.4	23	13.8	317.4	72	3.85	277.2	87.3
900	0.35	23.5	13.8	324.3	72.3	3.97	287.03	88.6
1000	0.2	25	14	350	75	4.24	318	90.82
1100	0.28	25.1	14	351.4	75.3	4.21	317	90.3
1200	0.3	25.2	14	352.8	75.5	4.23	319.4	90.5

Table 6 Variations in Duty Ratio during Open-Loop Analysis

Duty cycle	v_{in} (V)	i_{in} (A)	P_{in} (W)	v_{out} (V)	i_{out} (A)	P_{out} (W)	η (%)
0.2	23	13.8	317.4	72.3	3.89	281.25	88.6
0.4	23	13.8	317.4	72.8	3.88	282.46	88.9
0.6	25	14	350	75	4.25	318.8	90.56
0.8	25	14	350	75.3	4.19	315.5	90.2

$$B_{IASBC} = (B_{IASBC1} * d) + (B_{IASBC2} * d_1) \quad (28)$$

$$C_{IASBC} = (C_{IASBC1} * d) + (C_{IASBC2} * d_1) \quad (29)$$

The duty ratios during the on period of the switches and diodes are denoted by d and d_1 , respectively. Since the preferred converter operates in CCM mode, stage I correspond to the switch-on time, while stages II represent the switch-off time, as explained in section II. By considering these factors, we can derive the final averaged state matrix, which is expressed as follows in Eqs. (30) and (31).

This matrix captures the overall behavior of the system by incorporating the effects of duty ratios and CCM operation, providing a comprehensive representation for further analysis and control design.

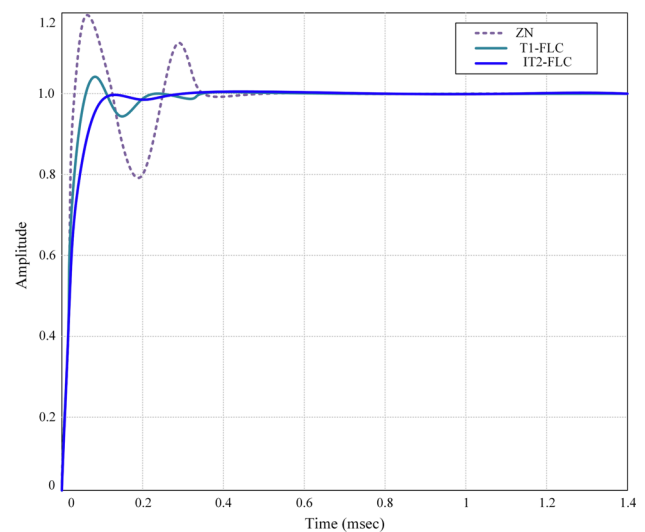
**Fig. 18** Step response for ZN, T1FLC and IT2FLC

Table 7 Reference Voltage Variations

$V_{ref}(V)$	$V_i(V)$	$i_i(A)$	$V_0(V)$	$i_0(A)$	η (%)
65	25	14	65	4.92	91.37
70	25	14	70	4.8	96
75	25	14	75	4.6	98.57

Table 8 Dynamic load variations for the proposed converter

Load (%)	v_i (V)	i_i (A)	v_0 (V)	i_0 (A)	Efficiency (%)
20	24.8	2.8	75	0.8	86.3
40	24.8	5.6	75	1.64	88.7
60	25	8.4	75	2.62	93.6
80	25.2	11.2	75	3.64	96.7
100	25	14	75	4.6	98.57

Table 9 Comparing the performance of the proposed converter to existing topologies

Parameters	Boosting Topology [37]	SEPIC [39]	Parallel switched-capacitor (SC) and Boost Converter [40]	Interleaved SC Multiplier [41]	Proposed Methodology
Voltage Gain	2	2	4	2	3
Diode count	1	1	5	4	2
Switch Count	1	1	5	4	1
Voltage stress of Switch	V_{in}	V_{in}	$V_{out}/2$	V_{in}	$V_{out}/2$
Controller	PSO	GA	PI	PI	IT2FLC
Efficiency	94.62%	95%	95%	95%	98.57%

Table 10 Comparison of various controller performances with the proposed controller based on its time domain specifications

Parameters	Open Loop	PI	PSO	GA	T1FLC	IT2FLC
Delay time (Sec)	0.11	0.06	0.04	0.03	0.015	0
Rise time (Sec)	0.2	0.212	0.18	0.16	0.12	< 0.04
Peak time (Sec)	0.5	0.4	0.17	0.16	0.131	0.1
Settling time (Sec)	0.6	1.3	1.2	1	0.8	< 0.1
Efficiency (%)	90.82	91.23	92.8	93.6	96.5	98.57

$$\frac{d}{dt} \begin{bmatrix} i_L \\ i_{L_A} \\ i_{L_B} \\ v_{C_A} \\ v_{C_B} \\ v_{C_L} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \frac{d_1}{L} & \frac{-C_B d}{L} & 0 \\ 0 & 0 & 0 & \frac{-C_A d - d_1}{L_A} & 0 & \frac{d_1}{L_A} \\ 0 & 0 & 0 & \frac{d_1}{L_B} & \frac{-C_B d - d_1}{L_B} & \frac{d_1}{L_B} \\ 0 & \frac{d}{C_A} & 0 & 0 & 0 & 0 \\ \frac{d+d_1}{C_B} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{-(d+d_1)}{R C C_L} \end{bmatrix} \times \begin{bmatrix} i_L \\ i_{L_A} \\ i_{L_B} \\ v_{C_A} \\ v_{C_B} \\ v_{C_L} \end{bmatrix} + \begin{bmatrix} \frac{d+d_1}{L} \\ 0 \\ 0 \\ 0 \\ \frac{d+d_1}{L} \\ \frac{d+d_1}{L} \end{bmatrix} [V] \quad (30)$$

$$V_0 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & (d + d_1) \end{bmatrix} \quad (31)$$

In this context, the state matrix is denoted as A_{IASBC} , the input matrix as B_{IASBC} , and the output matrix as C_{IASBC} .

These parameters play a significant role in characterizing the system dynamics and capturing the relationships between the state variables and the inputs/outputs. Each parameter contributes to specific aspects of the system's behavior, and their values can be adjusted to achieve desired performance or stability properties. The transfer functions that describe the overall performance of the method is ultimately expressed as follows.

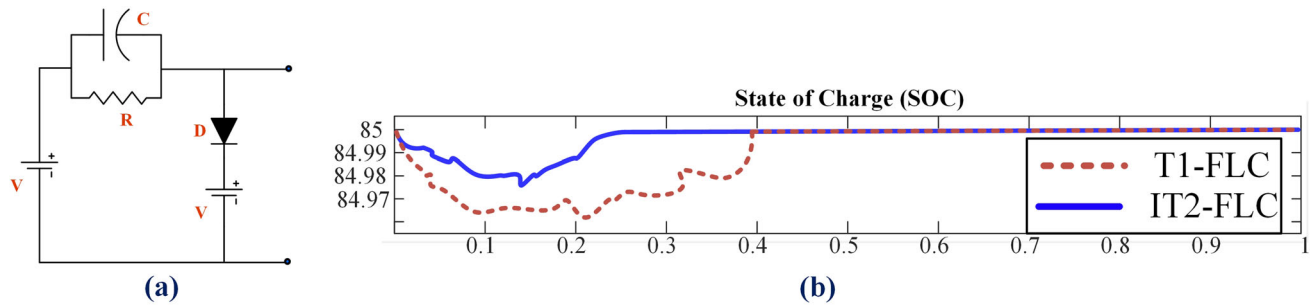


Fig. 19 **a** RC Battery Model **b** State of Charge

This transfer function captures the relationship between the input and output of the system and provides valuable insights into its dynamic response. By analyzing the transfer function, we can understand how the system processes and transforms inputs to generate the corresponding outputs. This understanding is crucial for designing appropriate control strategies, analyzing stability, and optimizing system performance.

$$G(s) = \frac{V_0}{V} = C_{IASBC}(SI - A_{IASBC})^{-1}B_{IASBC} \quad (32)$$

After taking into account the standards quantified in Table 3, the transfer functions derived from Eq. (33) can be expressed as follows:

$$G_{OL_IASBC}(S) = \frac{3.333e^4 S^5 + 4.22e^{-7} S^4 + 4.729e^{13} S^3 + 566.2 S^2 + 3.151e^{19} S}{S^6 + 1518 S^5 + 1.419e^9 S^4 + 2.154e^{12} S^3 + 9.455e^{14} S^2 + 1.435e^{18} S} \quad (33)$$

After obtaining the values of K_p and K_i as 1.272 and 0.268, respectively, through the Ziegler-Nichols method, the closed-loop transfer function with unity feedback can be derived as follows in Eq. (34):

$$G_{CL_ZPS}(S) = \frac{4.24e^4 S^6 + 8932 S^5 + 6.016e^{13} S^4 + 1.267e^{13} S^3 + 4.009e^{19} S^2 + 8.446e^{18} S}{S^7 + 4.391e^4 S^6 + 1.419e^9 S^5 + 6.231e^{13} S^4 + 9.582e^{14} S^3 + 4.152e^{19} S^2 + 8.446e^{18} S} \quad (34)$$

The bode plot shown in Fig. 7 illustrates the stability characteristics of the preferred system. It is evident from the plot that the system possesses a gain margin of 20.2 dB at a crossover frequency of $6.23e + 04$ rad/s, indicating a sufficient level of stability. Additionally, the phase margin of 92.4 deg at a crossover frequency of $3.77e + 04$ rad/s further confirms the system's stable behavior. These margins provide a measure of the system's robustness against uncertainties and disturbances. Overall, the bode plot analysis assures that the preferred system exhibits desirable stability properties.

5 Implementation of the proposed topology in MATLAB: a closed-loop approach

The suggested converter was analyzed using MATLAB Simulation software. The proposed circuit's design details are listed in Table 3. The solar panel delivers 25 V DC and 14A DC power as inputs to IASBC at 1000 W/m^2 irradiation and 25°C temperature, boosted to 75 V DC, 4.6A DC with a gain value of three. Figure 8 shows how the switches are actuated by a PI controller using double distinct suggested algorithms for multi-objective functions: T1FLC and IT2FLC. The ideal K_p, K_i, K_d and C solutions could be found via reducing the desired and rippling output voltages integral square error values. The PI controller gain parameters can be

adjusted through the proper selection of mutation constant and population size.

5.1 Pre-defined values for various algorithm types are set in advance

The PI controller that T1FLC and IT2FLC use to achieve multi-objective optimum tunes the ideal gain parameters K_p, K_i, K_d . Table 4 compares the performances of recognized and new algorithms kinds with IASBC. The parameters such as K_p, K_i, K_d , rise, peak interval and steady-state error for T1FLC and IT2FLC were obtained from Figs 9, 10, 11, respectively. Figure 12 contains a comparison chart between

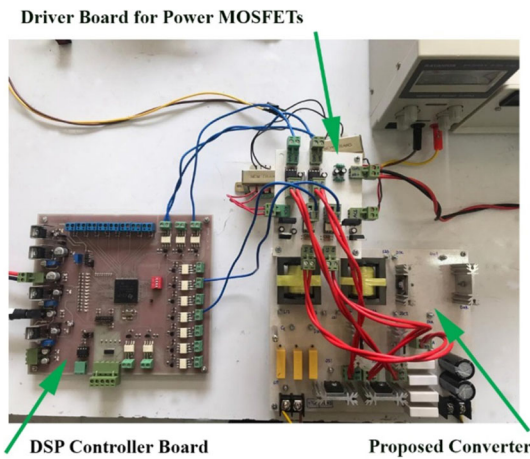


Fig. 20 Hardware configuration of the suggested IASBC converter

algorithms type and efficiency at $1000\text{W}/\text{m}^2$. Comparisons of proposed converter topologies performances utilizing identified and proposed techniques. Also, here the proposed IT2FLC controller provides an efficiency of 98.57%, which is comparably high compared to T1FLC (95.72%) and conventional ZN-PI controller (85.92%). The efficiency

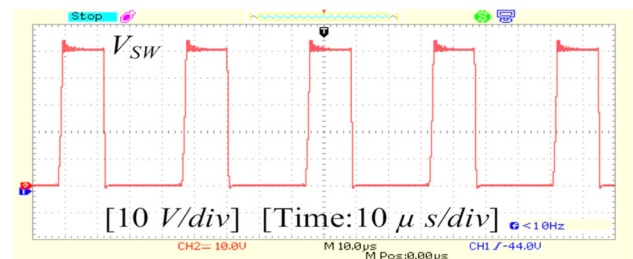
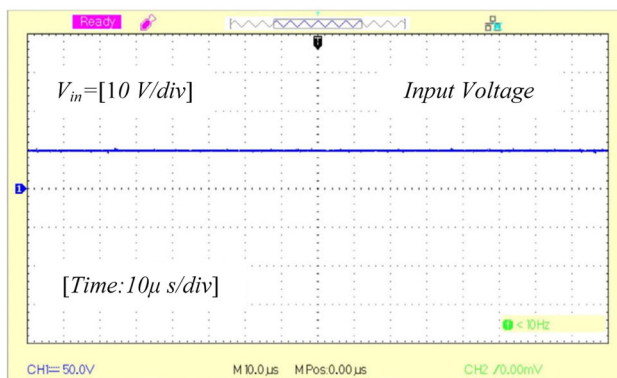


Fig. 22 Gating signal triggered by IT2FLC controller of 0.2 duty cycle

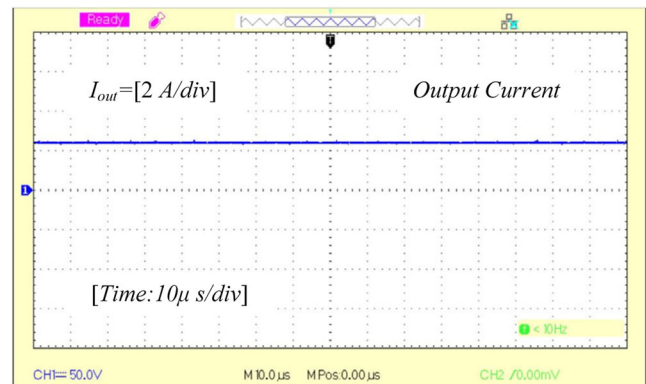
obtained is due to the optimal tuning of the PI gain parameters.

In Fig. 13, a consistent temperature of 25°C is depicted alongside time-varying irradiation. Figure 14 illustrates the utilization of the incremental conductance technique, an MPPT algorithm employed to optimize power allocation to loads and improve load management.

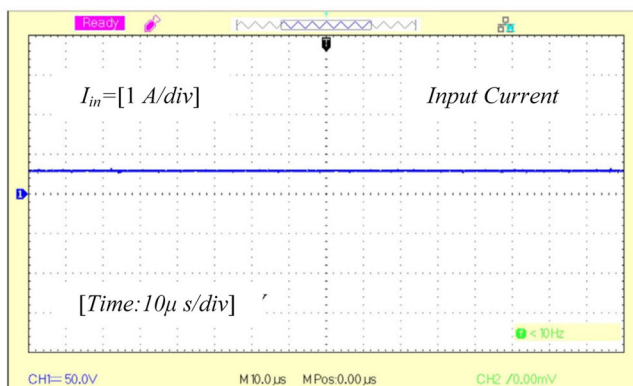
The PV and IV characteristics produced from the SPV after the MPPT are incorporated are shown in Fig. 15a. The panels deliver 350 W of power to the IASBC at 25 V DC and 14 A DC.



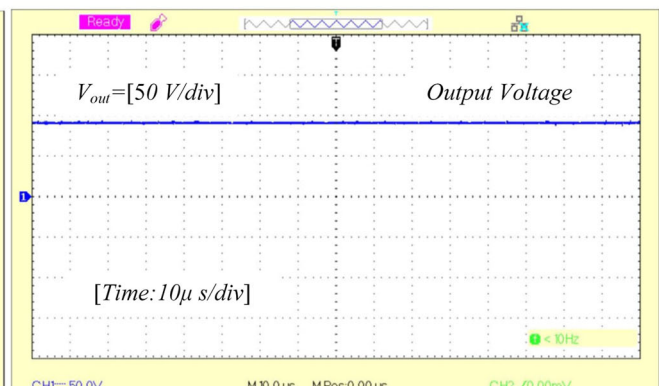
(a)



(c)



(b)



(d)

Fig. 21 Hardware outcomes **a** input voltage, **b** Input current, **c** Output current and **d** output voltage

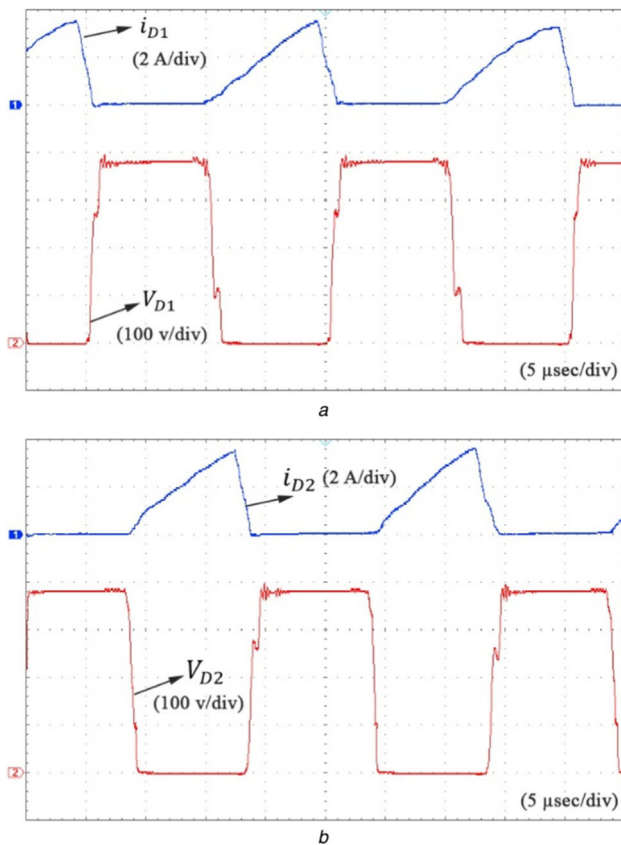


Fig. 23 (a) Current and Voltage characteristics of Diode **a** Diode D_A and **b** Diode D_B

5.2 Investigation and MATLAB Simulation Outputs of the Proposed Converter

The objective of this research is to enhance the responsiveness of the Intelligent Autonomous Solar Battery Charger (IASBC) by incorporating a feedback mechanism utilizing a PI controller-built MPPT techniques and tuning with T1FLC and IT2FLC. The employed approach for MPPT is the incremental conductance technique. In Fig. 15b, the MPPT converter's input voltage and input current are illustrated. To showcase the adjustment of the input voltage from 25 to 75 V through MPPT and T1FLC/IT2FLC controllers, Figs. 9a, 13b, 15a, and b are presented. Figures 9a and b demonstrate the output voltage of 75VDC and the current of 4.5ADC when the proposed topology is tuned with the T1FLC controller, revealing noticeable peak overshoot percentage and delayed settling intervals. However, by employing the IT2FLC controller, the system exhibits a rapid rise time of 0.02 s, a settling time of 0.025 s, and zero steady-state error, as depicted in Fig. 11a and b, respectively. Figure 11a and b showcase the steady-state output voltage and charging current waveforms of the IASBC employing the proposed MPPT and IT2FLC controller. The output response

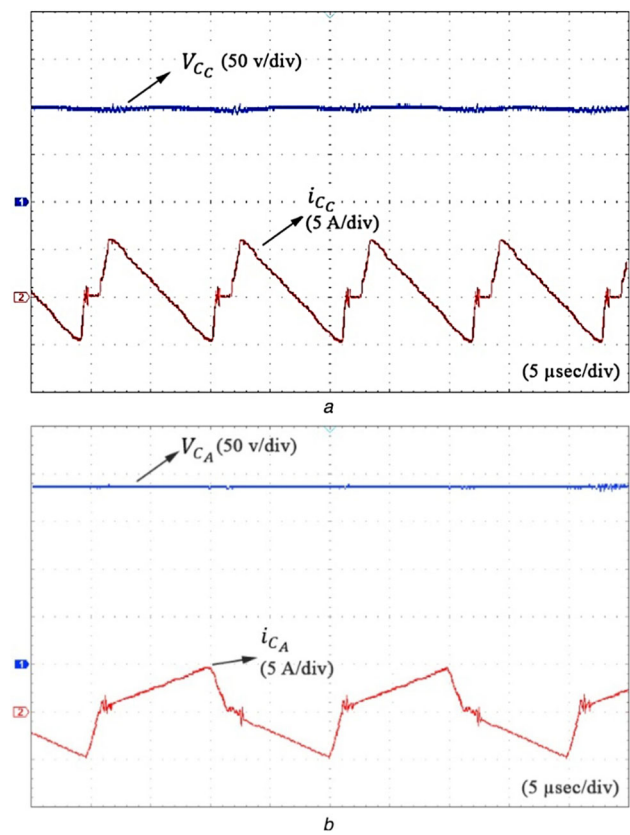


Fig. 24 (a) voltages (V_{CB}) and current (I_{CB}) waveforms of capacitors C_B (b) Voltages (V_{CA}) and Current (I_{CA}) waveforms of capacitors C_A

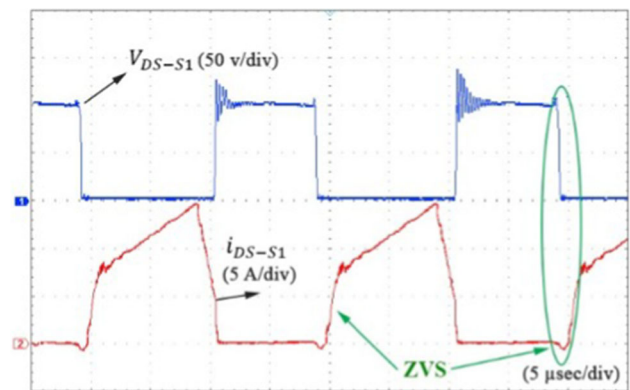


Fig. 25 switch S is triggered ON and OFF, the responses of drain-source voltages and current

of the IASBC is compared to that of the MPPT, T1FLC, and IT2FLC PI controllers under different tuning settings. The IT2FLC approach demonstrates superior performance with lower prior damping compared to the alternative controller. Figures 16 and 17 illustrate that the ripple values for T1FLC and IT2FLC are 0.6 V and 0.045 V, respectively. Consequently, the IT2FLC method effectively minimizes the ripple level in the output response. While the T1FLC approach

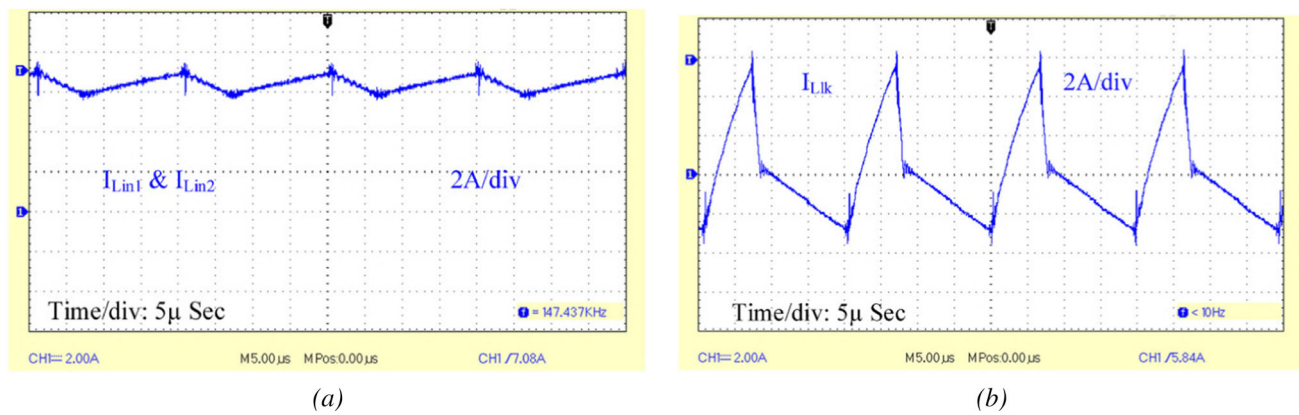


Fig. 26 **a** Coupling inductors and **b** primary inductors (I_L) exhibit different characteristics throughout charging and discharging

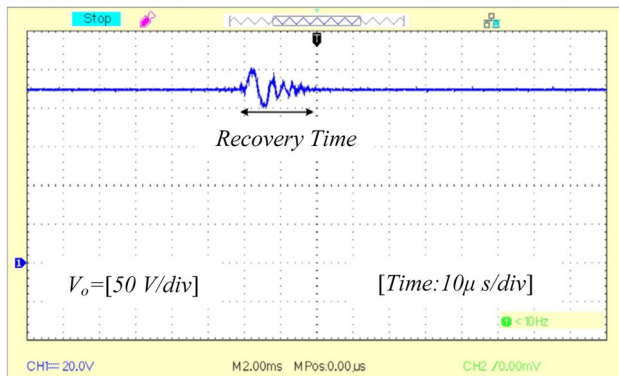


Fig. 27 Dynamic load change response of the proposed converter

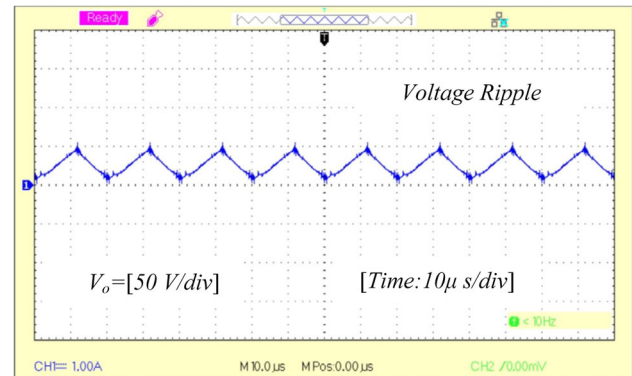


Fig. 28 Voltage ripple content of the output response

exhibits oscillations between 75.3 and 74.7, resulting in a ripple of 0.6 V, the IT2FLC technique shows oscillations between 75.035 and 74.99, resulting in a significantly lower ripple of 0.045 V. To assess the dynamic performance of an SPV-interconnected IASBC with an output voltage of 75 V, the input voltage is increased from 25 to 45 V at $t = 0.4$ s and then restored at $t = 0.45$ s, as depicted in Fig. 17. Additionally, at this time, the load is changed from 8.23 ohms to 7.23 ohms. Based on the simulation results of the IASBC, it is evident that the system yields several advantages, including higher output power and reduced peak overshoot in the input voltage. In Table 5, we observe an exploration of changes in irradiance patterns during an open-loop examination. Furthermore, Table 6 presents an examination of variations in the duty ratio throughout an open-loop analysis.

Figure 18 displays the step responses of the ZN-PI, T1FLC, and IT2FLC algorithms. The second plot illustrates the magnitude versus time, revealing that the IT2FLC approach outperforms the other two methods in terms of settling time, peak overshoot, steady-state error, and transients. Moreover, in the demonstration of IT2FLC with the

IASBC, an analysis of reference voltage variations is conducted to determine the optimal efficiency. The findings of this analysis are summarized in Table 7. Table 8 provides an analysis of dynamic load variations for the proposed converter, showcasing the system's adaptability to different load conditions. Table 9 presents an evaluation of the proposed converter's performance compared to existing topologies, highlighting its advantages and distinctive features. Lastly, Table 10 assesses the performance of different controllers in comparison to the proposed controller, considering their time domain specifications, and identifies the strengths and weaknesses of each controller.

In Fig. 19a, a battery model is simulated using MATLAB and Fig. 19b displays a state of charge characteristics obtained when using the controllers T1FLC and IT2FLC respectively where type 2 provides better and smooth charging characteristics than the type 1 controller. The study involves the computation and analysis of the equivalent impedance of this specific lithium battery type, which is then treated as a load consisting of a resistor and a capacitor.

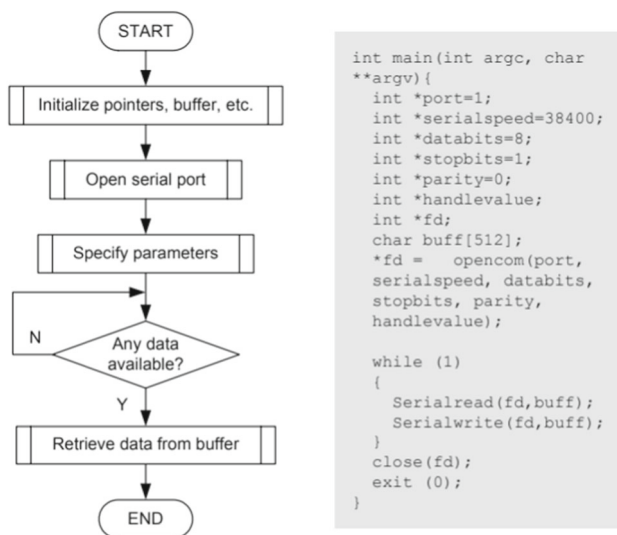


Fig. 29 Sequence stage of an embedded control algorithm

6 Experimental results of IASBC

The hardware configuration for the proposed work, aiming to achieve an optimal solution for the system elements through the implementation of MPPT and IT2FLC-based PI method, is illustrated in Fig. 20. The hardware components utilized in this setup include an Optocoupler TLP250, which is responsible for signal isolation and conversion from digital to analog. The dsPic30f2010 Controller is employed for overall system control, while IN4007 Diodes and IRFP460 MOSFET switches are utilized as well. In Fig. 21a, b, c, and d, the input voltage and current responses from the solar PV, operating at 25VDC and 3.956ADC, are presented. These responses are observed when the PV system is connected to the specified load condition through the converter. Additionally, the figure showcases the output voltage and current responses at an optimized state, achieving a high gain value of 75VDV and 1.26ADC. This optimization is achieved through the implementation of the IT2FLC-based optimization technique. As a result, the proposed system provides an optimal solution with a remarkably low ripple value of 0.046 V. The suggested controller method utilizes gating signals with a duty cycle of 0.2 to fine-tune the switch S in the proposed topology, as demonstrated in Fig. 22. In Figs. 23a and b, the voltage

and currents responses of diodes D_A and D_B are displayed, respectively. As the switches S is turned on and off utilizing the optimized gating pulse, the capacitors C_B and C_A charge and discharge accordingly. The voltages V_{CB} and currents I_{CB} obtained from capacitors C_B are depicted in Fig. 24a, while the voltage V_{CA} and current I_{CA} from capacitor C_A are shown in Fig. 24b. Figure 25a and b illustrates the response of the semiconductor switches S are turned on and off, and the current and voltage across the drain-to-source experience changes. To achieve high gain operation, a double-winding coupled inductor is employed, and its charging and discharging characteristics are depicted in Fig. 26a.

The charging and discharging characteristics of the source inductor L result in a significant increase in the source voltage, reaching higher gain values when the switches S is simultaneously triggered ON and OFF. This behavior is clearly illustrated in Fig. 26b. Figure 27 provides a visual representation of the dynamic load change response of the proposed converter, showcasing its ability to handle sudden additional loads effectively. Additionally, Fig. 28 presents the ripple content of the output voltage, demonstrating the stability of the converter's performance. Both the simulation and implementation of the proposed converter yield practically identical output results, indicating the high level of accuracy and consistency achieved. This confirms the effectiveness of the architecture in delivering optimal performance. To ensure optimal loading performances, a PI controller's technique is applied in the DSP controller panel, allowing for precise tuning of the PI parameters in the proposed power converter. The embedded control algorithm's sequential stages can be observed in Fig. 29, providing insights into the control process. In Table 11, a comparative analysis is presented, evaluating the efficiency of software and hardware implementations. This analysis allows for a comprehensive understanding of the advantages and drawbacks of each approach, aiding in the selection of the most suitable implementation method.

7 Conclusion

In conclusion, this research introduces a unique and innovative approach to enhance the performance of electric

Table 11 Efficiency comparison of Software and hardware implementation

Software Implementation (350W)						Hardware Implementation (100W)		
Open-Loop System			Closed-Loop System					
P_i (W)	P_o (W)	η (%)	P_i (W)	P_o (W)	η (%)	P_i (W)	P_o (W)	η (%)
350	318	90.82	350	345	98.57	98.9	94.5	95.55

vehicle battery charging systems. By combining a solar PV-integrated dc to dc boost converter, MPPT, and IT2FLC techniques, the proposed Improved A-Source Boost Converter (IASBC) overcomes the limitations of traditional converter topologies and optimization algorithms. The proposed system effectively charges a 48 V rechargeable battery within standard system parameters, with a voltage stability transition from 25 DC to 75 V DC. Experimental validation of the 100 W power-rated design confirms the technical significance of the proposed converter. Through comprehensive performance comparisons, it is evident that the incorporation of the IASBC and MPPT-based IT2FLC controllers significantly enhances load side performance, conversion efficiency, and voltage gain. The optimization technique employed in this study further reduces the output side dc ripple content, resulting in improved system performance. The software analysis reveals an efficiency of 90.82% in the open-loop system and 98.57% in the closed-loop system. Additionally, the hardware system achieves an efficiency of approximately 98.55%, thanks to the reduction in output ripple content through the proposed controller. The stability of the converter is evaluated through state space representation and confirmed by bode plot analysis, demonstrating its stability. Future research can focus on integrating advanced control techniques and optimization algorithms to achieve even higher levels of efficiency and performance. Furthermore, the application of the proposed converter design can be expanded to incorporate other renewable energy sources, providing sustainable charging solutions for electric vehicles. These avenues of investigation hold promise for further advancements in electric vehicle battery charging systems.

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Declarations

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